

Question Paper

Exam Date & Time: 19-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: ELE 1003/ELE_1003 FUNDAMENTALS OF ELECTRICAL ENGINEERING

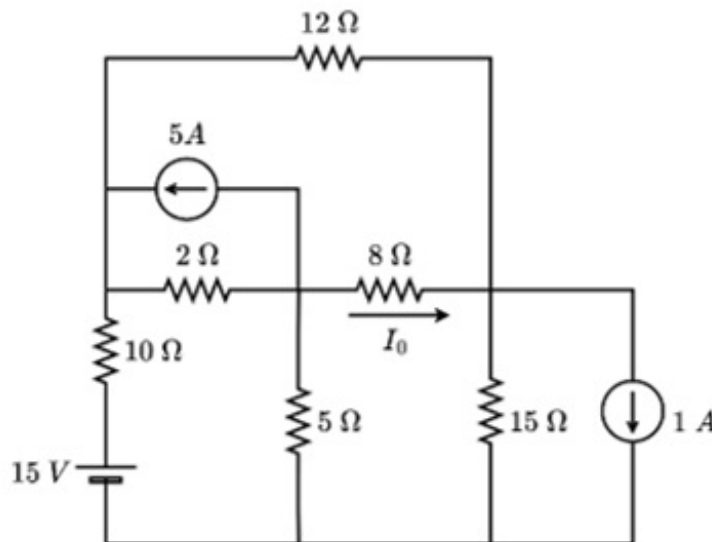
Fundamentals of Electrical Engineering [ELE 1003]

Marks: 50

Duration: 180 mins.

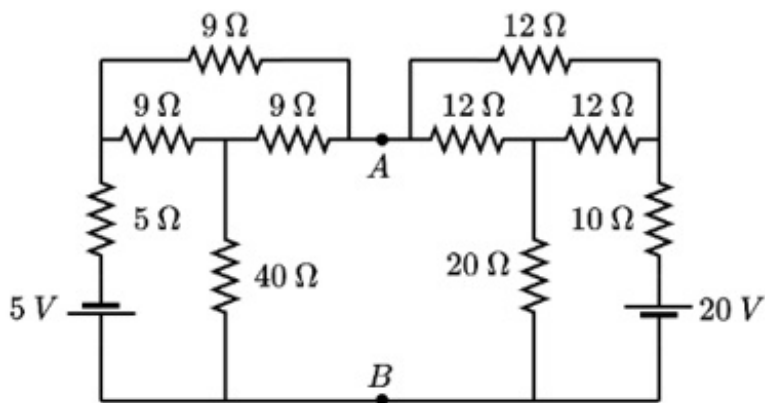
Answer all the questions.

- 1A) For the circuit shown in the figure, determine current I_0 using mesh current analysis. Also determine the power delivered by 15V source. (5)



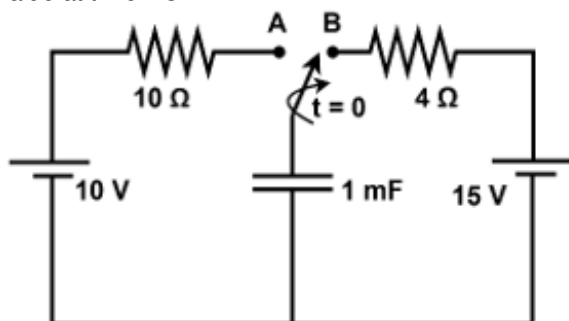
Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) Find Thevenin's equivalent of the network across terminals A and B, what is the value maximum power that can be transferred to variable load connected across A&B. (5)



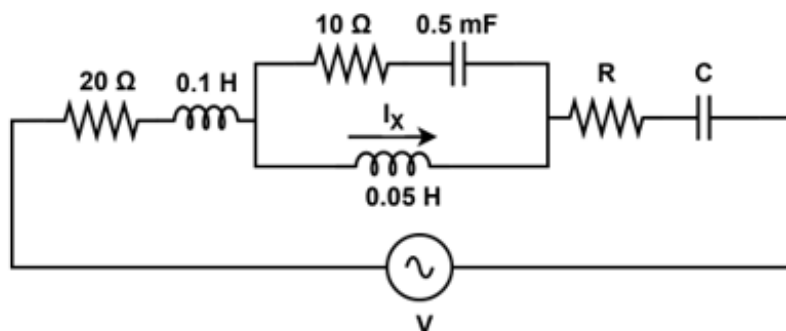
Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) The circuit shown below has achieved steady state when the switch is in position A. At $t = 0$ s, switch is moved to position B. Write the expression for voltage across the capacitor for $t > 0$ and find its value at $t = 5$ ms. (4)



Course Outcome: "CO1" / Cognitive Level: "Application"

- 2B) The circuit shown below has a 230 V, 50 Hz source. If the current through a 0.05 H inductor is $3.5 \angle -80^\circ$ A, taking supply voltage as the reference, (6)
- Find the value of 'R' and 'C'.
 - Determine the power factor of the circuit.



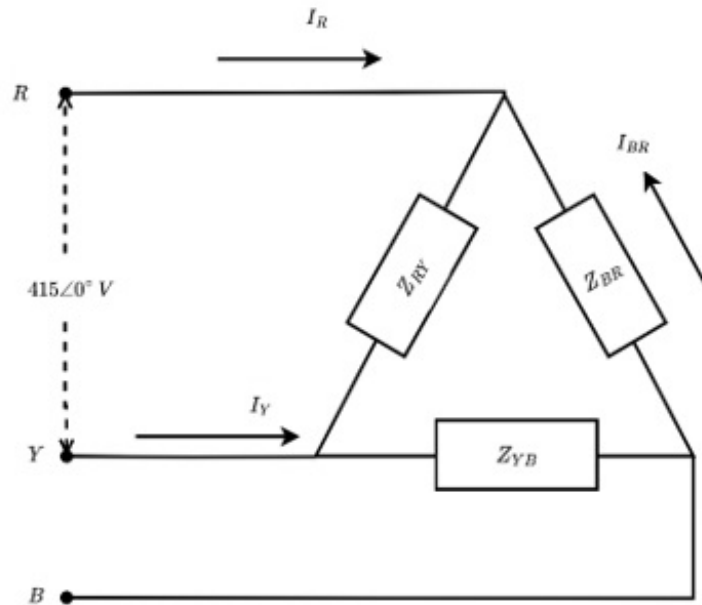
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) The following loads are connected in parallel across a 230 V, 50 Hz, single-phase ac supply. (4)
- 250 VA, 0.5 p.f lagging
 - 300 VA, 100 VAR (inductive)
- Obtain the complete power triangle.
 - Calculate the capacitance required in parallel with the loads to raise power factor = 1.
 - Both the loads are ON for 8 hours a day over a 30-day month. If the energy tariff is ₹6.00 per kWh. Calculate the monthly energy charges.

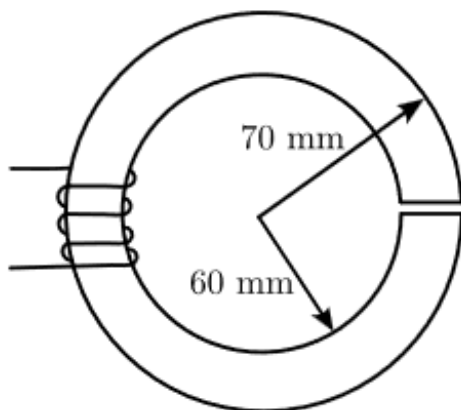
- 3B) A 3-phase, 415 V, RYB, 50 Hz balanced AC supply is connected to an unbalanced delta connected load as shown in the circuit. (6)

If $I_R = 5\angle -45^\circ \text{ A}$, $I_Y = 10\angle -30^\circ \text{ A}$, and $I_{BR} = 5\angle -60^\circ \text{ A}$ then

- Find the value of all the impedances in ohms.
- Total power consumed by the load. Sketch the phasor diagram clearly showing line voltages, line & phase currents. Take V_{RY} as reference.



- 4A) A ring made of ferromagnetic material has a cross-sectional area of 400 mm^2 , with inner and outer radii of 60 mm and 70 mm, respectively. The core is uniformly wound with a coil of 1000 turns. Determine the relative permeability of the ring if a current of 5 A produces a flux of 1 mWb in an air gap of length 2 mm. (4)



- 4B) What are the different losses that occur in an induction motor? Why are starters necessary for induction motors? Explain why single-phase induction motors are not self-starting. (3)

- 4C) Explain any three renewable energy systems used for power generation, highlighting their key features and components. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5A) List the main components of a digital measuring instrument and describe their functions. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5B) Compare a Linear Power Supply and a Switched Mode Power Supply (SMPS) in terms of their design, efficiency and applications. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) List the classification of power electronic converters and briefly explain one example in each. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ELE_1003:FUNDAMENTALS OF ELECTRICAL ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	16
3	3	CO 3	7
4	4	CO 4	6
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ELE_1003:FUNDAMENTALS OF ELECTRICAL ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	16(32%)
3	Application	4	18(36%)
4	Analysis	3	16(32%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)